



ESP32 to MAX4820 connections (part 2)

1 message

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Here is a textual version of the same (worked out after a long session with Claude and all the data sheets). I've checked the work against the data sheets and I think we're all good. I had to give Claude images of the chip to get the pinouts correct. He's not very good at getting the pinout right from the PDF data sheets.

COMPLETE CONNECTION TABLE: ESP32-S3-DevKitC-1 to MAX4820

SPI and Control Signals:

Function	ESP32-S3 Pin Name	GPIO#	DevKit Header Pin	MAX4820 Pin #	MAX4820 Pin Name	Notes
SPI Clock	FSPICLK	GPIO12	J1-18	6	SCLK	Connect to all 5 MAX4820s
SPI MOSI	FSPID	GPIO11	J1-17	5	DIN	Connect to all 5 MAX4820s
Chip Select 1	GPIO10	GPIO10	J1-16	4	CS	MAX4820 #1 only
Chip Select 2	GPIO13	GPIO13	J1-19	4	CS	MAX4820 #2 only
Chip Select 3	GPIO14	GPIO14	J1-20	4	CS	MAX4820 #3 only
Chip Select 4	GPIO21	GPIO21	J3-18	4	CS	MAX4820 #4 only
Chip Select 5	GPIO1	GPIO1	J3-4	4	CS	MAX4820 #5 only
SET (optional)	GPIO2	GPIO2	J3-5	2	SET	Connect to all 5 MAX4820s with 10kΩ pull-up
RESET (optional)	GPIO42	GPIO42	J3-6	3	RESET	Connect to all 5 MAX4820s with 10kΩ pull-up

Power and Ground (All 5 MAX4820s):

MAX4820 Pin #	Pin Name	Connect To	Notes
1	V_CC	3.3V or 5V	Logic supply, add 0.1μF bypass capacitor to GND
9	GND	GND	Ground
12	PGND	GND	Power ground for outputs
18	PGND	GND	Power ground for outputs
15	COM	V_CC or relay supply	Common cathode for kickback diodes (max 6V)
EP	Exposed Pad	GND	Connect exposed pad to ground plane

Relay Output Pins (Each MAX4820):

MAX4820 Pin #	Pin Name	Function	Connect To
20	OUT1	Output 1	Low side of relay coil #1
19	OUT2	Output 2	Low side of relay coil #2
17	OUT3	Output 3	Low side of relay coil #3
16	OUT4	Output 4	Low side of relay coil #4
14	OUT5	Output 5	Low side of relay coil #5
13	OUT6	Output 6	Low side of relay coil #6
11	OUT7	Output 7	Low side of relay coil #7

MAX4820 Pin #	Pin Name	Function	Connect To
10	OUT8	Output 8	Low side of relay coil #8

Unused Pins:

MAX4820 Pin #	Pin Name	Connection
7	DOUT	Leave unconnected (not daisy-chaining)
8	N.C.	No connection

Important Implementation Notes:

1. **PGND Connections:** Connect both PGND pins (12 and 18) together and to the main GND for proper current return path.
2. **Bypass Capacitors:** Place a 0.1 μ F ceramic capacitor between V_{CC} (pin 1) and GND (pin 9) on each MAX4820, as close to the IC as possible.
3. **SET/RESET Pull-ups:** Add 10k Ω resistors from SET (pin 2) and RESET (pin 3) to V_{CC} to keep them inactive (high) when not driven.
4. **COM Connection:** Can be connected to V_{CC} or to a higher voltage (up to 6V) for faster inductive kickback recovery. If using higher voltage, add 0.1 μ F capacitor from COM to PGND.